Radio Controlled Beach Cleaning Bot

Nasreen Bano
Department of Mechatronics Engineering
SZABIST
Karachi, Pakistan
nasreen.bano@szabist.edu.pk

Amina Amin
Department of Computer Sciences
DIHE
Karachi, Pakistan
aminaamin@dadabhoy.edu.pk

Hussain Boghani
Department of Mechatronics Engineering
SZABIST
Karachi, Pakistan
hussainboghani@gmail.com

Hiba Tariq
Department of Mechatronics Engineering
SZABIST
Karachi, Pakistan
hiba_tariq4@gmail.com

Saba Bakhtawar
Department of Mechatronics Engineering
SZABIST
Karachi, Pakistan
saba.bakhtawar987@gmail.com

Illyas Waggan
Department of Mechatronics Engineering
SZABIST
Karachi, Pakistan
illyaswaggan@gmail.com

Tanzila Younas
Department of Mechatronics *Engineering*
SZABIST
Karachi, Sindh
tanzila@szabist.edu.pk

***Abstract*— the pollution, now a days occurring on beaches, is increasing day by day due to some scathing events taking place such as flood or solid litter left behind by the public. This litter, when present in a large quantity of small pieces, appears to be more hazardous for the animals, both inside and outside of the water. Cleaning this litter requires a large no. of labors that add to the cost. To overcome this obstacle, there have been many systems designed on a larger scale, each differing in quantity and type of litter as well as the structure. A system that can work efficiently in coastal environment and is cost effective, reducing human effort and wastage of time, is still the need of the present day. Our project is one suitable method to reticulate this issue. The prototype of RC beach cleaning bot is designed and fabricated for this purpose. It comprises of filtration mechanism and driving mechanism. The filtration mechanism consists of a system used for the separation of sand and litter of small size (i.e. plastic pieces, glass pieces, cans, cigarette butts etc.) using vibration mechanism. Whereas, the driving mechanism remotely controls the motion of the robot via Bluetooth module that will send and receive signals with the help of a transceiver. This type of system works significantly for the environment in best economic way.**

Keywords— ***Pollution, beaches, solid litter, hazardous, beach cleaning bot.***

I. INTRODUCTION

The environmental pollution is increasing rapidly now a days and one of the major contribution to this is poor condition of beaches. Where beaches on one hand add beauty to the countries, are steadily becoming sources of exploiting this natural beauty and negatively affecting the country [1-3]. To overcome this issue swiftly and in an economic manner, many beach cleaning systems were designed previously. Some of these systems were designed and fabricated on a larger while some on small scale, including systems that require human effort to drive them along the way. These are described below:

A. Garbage Collection Robot:

This robot, as the name suggests, is a garbage-collecting robot on beaches. It operates using a wireless connection. It also has an IP camera with an Ad-hoc system, which adds up in the cost. This robot can move with the velocity of 0.5 m/s. The power supplied from a 12V 30Ah battery is connected to 40W solar cell. It is not used to separating out tiny debris.

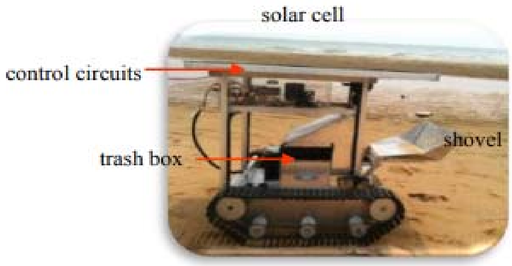

Fig. 1. Side view of garbage collection robot [4].

B. Beach Cleaning Trailer:

The beach cleaning trailer uses a conveyor belt to transfer trash into the shaking tray and then collect it in the trash bin. The shaking tray is used to extract associated trash from the sand. It is less sophisticated in functioning as it uses a hydraulic system for controlling and actuation purposes.

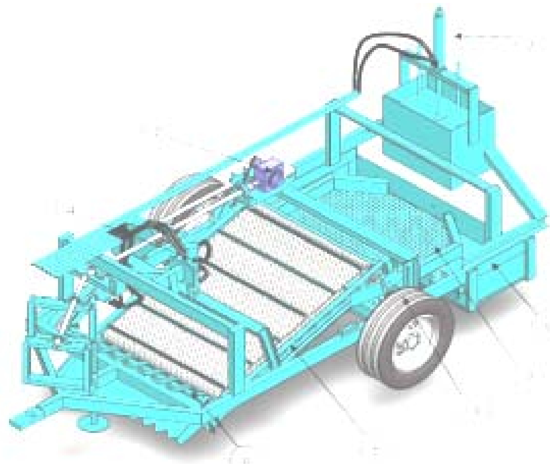

Fig. 2. Beach Cleaning Trailer [5].

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C. HS-GreenFist - Beach Cleaner Robot:

The robot that designed is capable to navigate on the beach and collect cans found in its surrounding. Its structure is fabricated using cheap materials i.e. steel, medium density fibre and acrylic, to meet the objective of making it light weight as well as cost effective. However, this can affect its durability. The design includes an arm, that allows the robot to pick up the cans, and a ramp that manages to drop them into the container. The movements of the arm and ramp are carried out by mean of servomotors. Kinect is also used to measure the depth. Above all features it have, it is not able to collect small trash such as, cigarette butts and plastic pieces, which are critical to marine life.

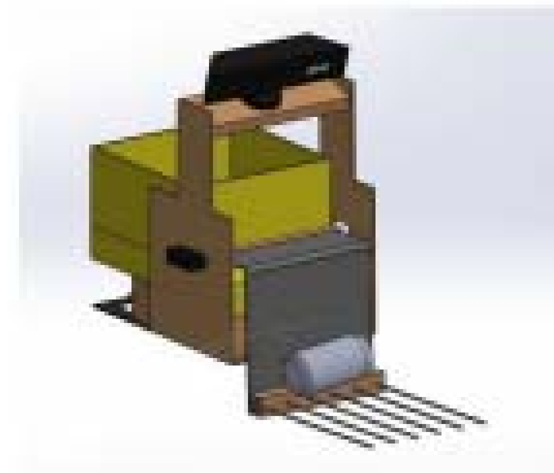

Fig. 3. HS-GreenFist- beach cleaner [6].

D. Prometeo – Beach Cleaner Robot:

Its objective is similar to that of the HS-Green Fist beach cleaning robot but operates with a different mechanical structure. It has an excavator arm that is designed in a way that it even carries out cans that half buried in the sand. To achieve a specific height in accordance with the excavator, a chain mechanism (differential wheeled) is used for its movement. A system is slightly complex because a 3D camera is implemented, with its algorithm, to achieve desired distance and position.

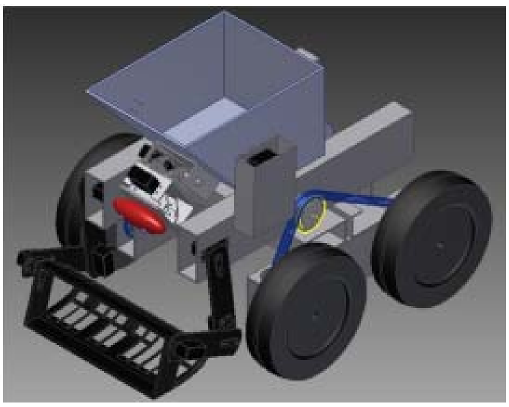

Fig. 4. Prometeo- Beach cleaner [7].

E. Modular Robot as a Beach Cleaner:

This robot is designed with a high degree of robustness and has features of trash collecting system, with the help of its excavator like claw, a closed loop motor speed control system and a computer vision system, to detect and avoid obstacles. It uses 3mm thick aluminium plate to form a robust system. A storage compartment collects all the trash separated. However, this robot is constrained to collect only cans and object similar to them, leaving around small trash.

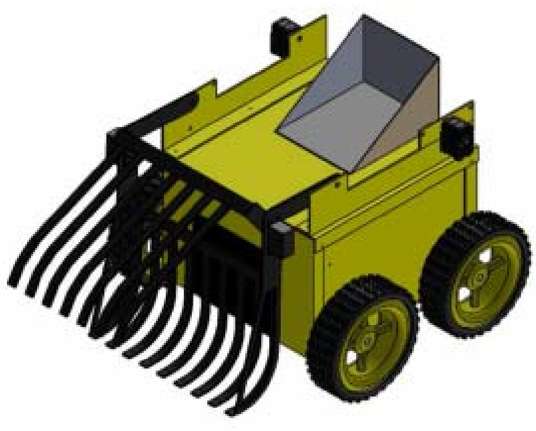

Fig. 5. Modular beach cleaner [8].

F. Hirottaro:

This prototype is an autonomous system and works with the refuse collection mechanism. It is based on the action of a broom and a dustpan. A navigation system is involved in it, which is used to calculate the exact position and orientation of the robot. It is bigger in size and weight is 220kg. It travels with a velocity of 0.65m/s. To operate on uneven sand a terrain, a brush is used which remains in contact with sandy terrains during its operation. The system is autonomous but at the same time is technically sensitive due to its increased components.

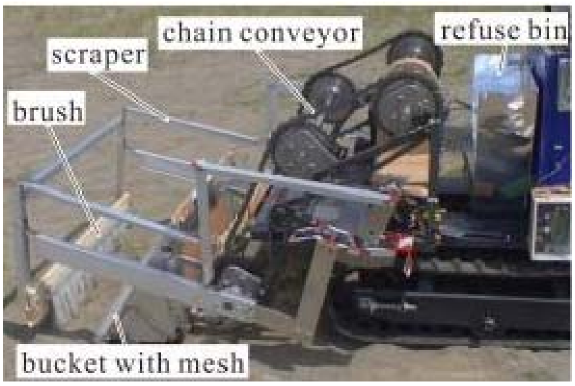

Fig. 6. Hirattaro beach cleaner [9].

One prime issue with these smaller versions is that they are not capable of collecting small debris from the beach sand, which are a major threat to animals. To design a radio-controlled robot, which has similar working features as that of an autonomous robot with more efficiency, is itself a challenging task. Our main aim is to work on and provide a system that not only is economic compared to other systems being used for the same purpose but also is capable of filtering out sand from small litter found on beach sand. In addition to that, it is also important to get maximum productivity out of it.

II. IMPORTANCE OF BEACH CLEANING SYSTEM

From recent studies, it is shown that the pollution that occur on beaches is restricted to local area and can take place due to the disturbance to wastewater collection or handling of infrastructure affected due to some critical events such as illegal dumping, floods or solid litter left behind by public. Materials like rubber, plastic, metal or foam are brittle enough i.e. hardly breakable, and are menacing to the life of animals, especially the marine animals. This kind of litter and trash becomes more savage when it is present in the form of small pieces. When small or big animals intake it, some get drown from attachment while others are clogged up by plastic rings. The deposits of this litter increase with every curling sea wave that breaks on the shore. This problem will continue to stand on its way if ignored, since it is a never-ending process. Since a lot of us eat seafood, so if the fish that we eat consumes that plastic piece then we are consuming that piece too [1]. This problem not only endangers marine life but also creates a negative impact upon tourism by destroying the natural beauty of the beaches. To eliminate this issue, a lot of machines and robots were designed previously but all of them work on a large scale and require extra labor expenses as well. Some of them were made on a small scale but are not capable of removing small debris and trash that is covered by heavy layer of sand. The thought of designing this project is of an innovation or invention of a radio-controlled system that contributes to the country's betterment and is the need of the present day. This radio-controlled beach cleaning bot will not only add to the country's growth in the field of technology but will also eliminate an inefficient and a long-term way of cleaning beaches i.e. by gathering number of people for the collection of small debris from coastal areas of the country, which is the major issue that is increasing day by day due to the lack of facilities and equipment [10-16].

III. RADIO CONTROLLED BEACH CLEANING BOT

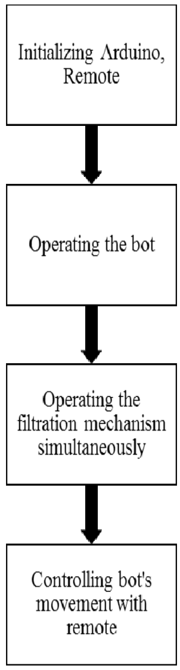

Fig. 7. Workflow diagram of RC beach cleaning bot.

To implement accordingly, beach cleaning bot was divided into parts, hardware and software. Hardware part is comprised of mechanical structure, which was further divided into two categories, one is filtration segment, which consists of filtration mechanism, and the other is the driving segment, which provides driving force to the bot as it contains driving mechanism in it. Whereas software part is comprised of CAD design and programming of the beach cleaning bot. The hardware is designed in a way that the bot will be capable of cleaning the sandy surface using its filtration mechanism and the driving mechanism will power it to simultaneously move with a continuous motion in a straight line on that surface. Below described is the filtration segment and the driving segment of the beach cleaning bot.

A. Filtration Segment

Filtration segment of the bot comprises of the filtration mechanism. This mechanism is similar to the four bar mechanism. It has a motor that rotates and a connecting bar linked to the motor, followed by a system that filters out sand by oscillating. This mechanism is structured with an iron frame to make it strong enough to bear the mechanical load easily on it. Below discussed are all the components that contribute in completion of this segment:

a) *Frame:* The structure of the filtration segment consists of an iron frame of 20 x 15in that holds all the other parts of the filtration segment in it. The iron metal chosen to construct the frame is due to the reason that iron serves the robust as well as less costly material as compared to aluminium and steel.

b) *Plough Arm:* A sand is ploughed up on the sieve by a plough arm. A plough arm consists of metal alloy sheet of size 3.5 x 3.5in. The arm of 17in in length that is joined to the frame at an angle of 30° from the ground. This angle is set to get the maximum sand ploughed out of the ground.

c) *Motor:* The filtration mechanism uses a 24V DC geared motor as a rotating body, which provides high torque and speed. It is connected with the shaft in a way that it tends the shaft to move in back and forth motion and that is how the mechanism produce vibrations.

d) *Shaft:* A shaft of length 20in in is the part of filtration mechanism. It is connected to the sieve in a way that it acts as a transmitter of an oscillatory motion to the sieve.

e) *Sieve:* One of the major part of the filtration mechanism is the sieve, which is designed and manufactured using a net and a frame of metal alloy of size 17 x 15in. For the purpose of the sieve to separate sand from small debris or litter ploughed from the surface properly, it contains a wire mesh of size 0.65, much smaller than the size of a bottle cap. The plough arm attaches with the sieve at an angle of 30° to transfer all the ploughed sand on it. The frame of the sieve makes it robust enough to withstand the force with which it moves in back and forth motion to filter out the sand and prevent the wire mesh from getting bended.

f) *Trash Storage Container:* On the backside of the frame is attached the storage container of size 20 x 6in. The storage container stores in it all the remaining small debris and litter collected on the surface of the sieve, separated from the sand. The storage container is 7in in height; therefore, it can store a limited amount of trash in it.

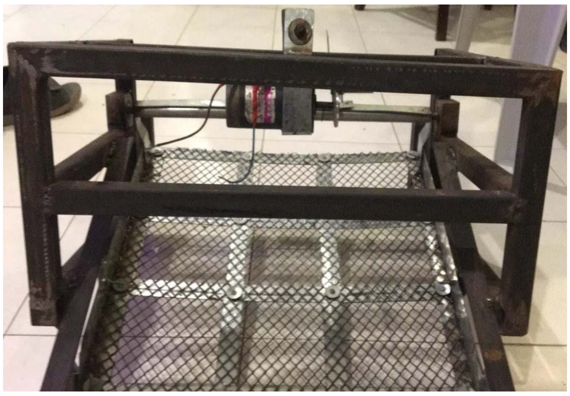

Fig. 8. Front view of filtration segment of the bot.

B. Driving Segment

This segment serves as the driving force of the project. It consists of the components that will drive the bot through, ensuring proper functioning of all the mechanisms at the same time.

In the driving segment of this bot is installed a driving mechanism that include motors with gearboxes that are joined with the tires to increase the torque, a relay based motor driver circuitry that provides more power to the tires, batteries that provides power to both the mechanism and last but not the least Arduino module that functions the bot. The mechanism is structured with iron frame to make it robust and metal sheets to provide a cover to it, preventing the internal body from outer atmosphere.

Fig. 9. Side view of driving segment of the bot.

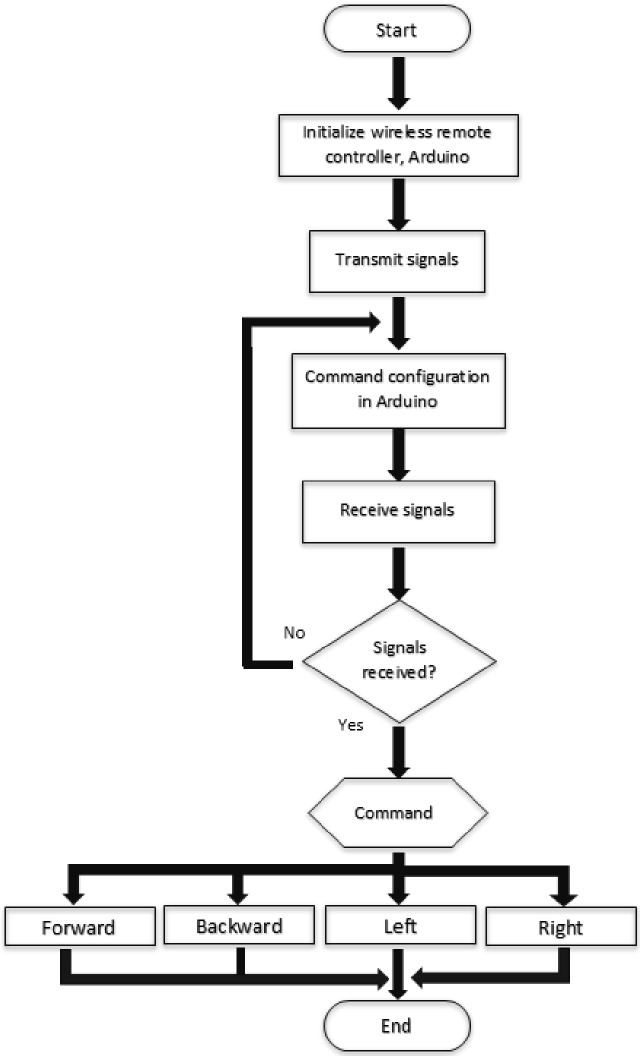

Fig. 10. Flowchart of driving mechanism.

To power up both filtering mechanism and driving mechanism we have used a combination of Li-Po batteries, which are able to provide current of three Amps.

IV. WORKING PRINCIPLE

A radio-controlled beach cleaning robot works in a way that when the user powers on the main switch, the system starts, enabling the Arduino microcontroller. The remote controller, which acts as a transmitter to Arduino module, is programmed in a manner that as it is switched on and gives command, either forward or reverse, the transmitter sends a signal to the receiver using radio waves. The receiver then delivers that signal to Arduino, which then drive the respective motors. This is how the bot moves and complete its traveling range of 1km. Now the working of filtration mechanism, in order clean the sand, is carried out in a way that as the switch attached to the DC motor gets high, the motor is operated. This motor tends a link, which connects with it the motor and the shaft, to rotate. The shaft joined with the sieve tends to move the sieve in to and fro motion (oscillatory motion). The sieve in this way produces vibratory motion. As the user drives the bot on the sand via remote controller, the plough arm of the filtration segment ploughs the sand simultaneously. The ploughed sand is then moved upwards up to the sieve, which then vibrates in order to filter out the sand. In this way, the sand falls through the mesh grid and the remaining litter in moved forward and fall in the trash storage container. This trash container collects litter up to 6in of height, which then be emptied easily, making space for the upcoming litter. Both the processes of driving and filtering are carried out simultaneously once the beach cleaning bot is powered on.

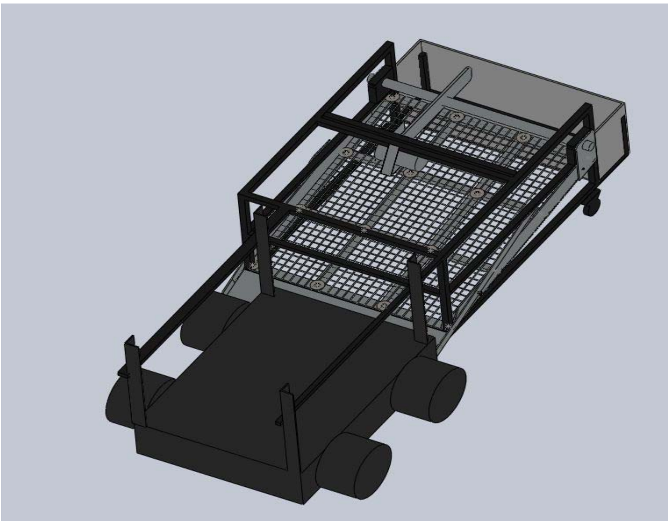

Fig. 11. CAD design of RC beach cleaning bot.

Fig. 12. RC beach cleaning bot in running mode.

Fig. 13. Side view of a beach cleaning bot.

V. CALCULATIONS AND ANALYSIS

To calculate the torque and power of driving and filtration units motors respectively:

a) Torque of Driving Motor:

Mass of filtration unit = 16kgs
Mass of driving unit = 14kgs
Total mass of the bot = 30kgs
Weight of the bot = W = 30kgs x 9.81m/s2 = 294.3N
Diameter of a tire = D = 6in = 6 x 2.54 = 15.24cm
= 0.1524m

For each tire one motor is used, to calculate the torque of a single motor divide the weight,

F = W/4 = 73.575N

Therefore,

$$Torque = F \times D \tag{1}$$

= 73.575 x 0.1524
= 11.212N.m

b) Power of Filtration Motor:

Speed = 800rmp
Mass = 1.5kg
Crank radius = r = 1in = 0.0254m
Force = 14.715N

$$V = \frac{2\pi}{60} \times r \quad (2)$$

From the above equation, velocity is found to be:
Velocity = 2.1279m/s

$$Power = F \times v \quad (3)$$

By putting the values in above equation, power is:
Power = 31.312W

TABLE I. CURRENT TO TIME.

Time (ms)	100	500	1000	1500
Current (A)	2.77	2.45	2.03	1.48

The above observations carried out to find the current required by the motors to function properly and the time it took to dissipate the current.

VI. DISCUSSION

A radio controlled beach cleaning bot overall worked but with the range of only 1Km. It can be moved in forward and reverse direction and can collect the trash in a container up to 7inches of height. The main factor in overall designing of the robot was its mechanical structure. The material selection, joints, dimensions and battery selection were also other important items of concern. All of this is done by keeping in mind the main goals of the project. The bot is made of iron metal with rust inhibitor, making it robust. It is economic and serves for great purpose of reducing environmental pollution.

VII. CONCLUSION

A radio controlled beach cleaning bot, which is designed with the motive to clean the sandy surface of the beach, is a good choice for reducing environmental pollution that occur due to large amount of litter found on the beach. Therefore, it gives you maximum productivity and consumes sufficient amount of power. It is easy to handle and less costly that makes it an efficient robot among others. The main advantage of using this bot as a beach cleaning system is that it replaces the need of so many labors to clean the beach with one single system and a user that remotely controls the bot, eliminating labor cost and saving time simultaneously.

REFERENCES

[1] Sirichai Watanasophon, Sarinee Ouitrakul, “Garbage Collection Robot on the Beach using Wireless Communications,” in International Conference on Informatics, Environment, Energy and Applications. , 2014.

[2] Younas, T., Akbar, M. A., Motan, I., Ali, Y., & Atif, M. (2017). Design and fabrication of an autonomous multifunctional robot for disabled people. In *2017 4th IEEE International Conference on Engineering Technologies and Applied Sciences (ICETAS)* (pp. 1-6). IEEE.

[3] Bano, N., Aziz, M., Ghani, U., Taha, M., & Ahmed, H. (2018, September). Automatic Book Scanner. In *IOP Conference Series: Materials Science and Engineering* (Vol. 417, No. 1, p. 012032). IOP Publishing.

[4] N.Kiran Kumar, G.Vivek, Dr.A.Pulla Reddy, “Garbage Collection Robot on the Beach Using Wireless Communications.,” International Journal of Research in Engineering, Science and Management. , vol. 1, 2018.

[5] Kuson Prakobkarn, Banyat Saitthiti, Sakda Intaravichi, “Design and Construction of Beach Cleaning Trailer by Finite Element Method,” vol. 3, 2012.

[6] Rel Guzman Apaza, Edwin Gutierrez Linares, Enrique A. Soto Mendoza, Elvis D. Supo Colquehuanca, “HS-GreenFist: Beach Cleaner Robot.,” in IEEE Open Category., 2013.

[7] Oscar Cieza, Carlos Ugarte, Elizabeth Gutiérrez, Javier García, José Tafur, “Prometeo: Beach Cleaner Robot .,” 2015.

[8] Felippe Schmoeller da Roza, Vinicius Ghizoni da Silva, Patrick Jose Pereira, Douglas Wildgrube Bertol, “Modular robot used as a beach cleaner,” Ingeniare Chilean Engineering Magazine, vol. 24, 2016.

[9] Tomoyasu Ichimura, Shin-ichi Nakajima, “Development of an Autonomous Beach Cleaning Robot-Hirottaro.,” in International Conference on Mechatronics and Automation August 7 - 10, Harbin, 2016.

[10] Dinkar R Patnaik Patnaikuni, “A Comparative Study of Arduino, Raspberry Pi and ESP8266 as IoT Development,” International Journal of Advanced Research in Computer Science, vol. 8, p. 3, 2017.

[11] Yuan Fu-cai, Hu Shi-jian, Sun Hai-liang, Wang Li-zhu, “Design of Cleaning Robot for Swimming Pools.,” in International Conference on Management Science and Industrial Engineering (MSIE), 2011.

[12] Waqar Farooq, Nehal Butt, Sameed Shukat, Nouman Ali Baig, Sheikh Muhammad Ahmed, “Wirelessly Controlled Mines Detection Robot,” in IEEE, 2016.

[13] Wei Chen, Yongguo Zhao, Chengye Liu, Mingming Jiang, Jie Sun, “Design and Implementation of Control System for Nuclear Pollution Disposal Robot based on Wireless,” in IEEE, Jinan, 2017.

[14] Jinyu Zhu, Chunlei Liu, Weixuan Sun, “A Remote Controlled Mobile Robot based on Wireless Transmission,” in 2nd IEEE Advanced Information Management,Communicates,Electronic and Automation Control Conference(IMCEC 2018), Shanxi, 2018.

[15] Akash Singh, Tanisha Gupta, Manish Korde, “Bluetooth controlled spy robot,” in IEEE International Conference of Information, Communication, Instrumentation and Control, Indore, 2017.

[16] Yusuf Abdullahi Badamasi, “The Working Principle Of An Arduino,” in IEEE, Abuja, 2014.